

Status: Posted



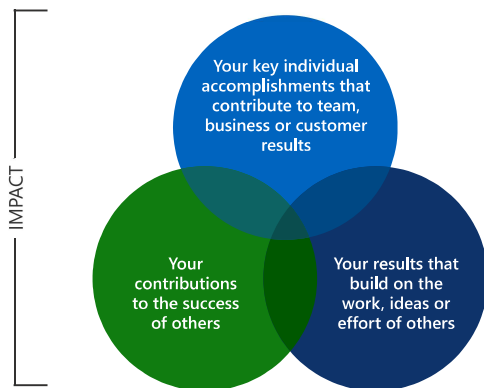
DEVESH KUMAR

SOFTWARE ENGINEER II | Reports to: Suma Kollipara

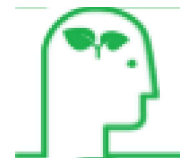
Connect Oct 2023

Reflection Period: Apr 30, 2023 - Oct 23, 2023

Connects provide a regular touchpoint where you and your manager can come together with a growth mindset to reflect on impact and discuss areas for improvement and future development.



Fixed Mindset leads to a desire to look smart and therefore a tendency to avoid challenges, give up easily, see failures as fruitless or worse, ignore useful negative feedback and feel threatened by the success of others.



Growth Mindset leads to a desire to learn and therefore a tendency to embrace challenges, persist in the face of setback, see failures as essential to mastery, learn from criticism and find lessons and inspiration in the success of others.

Reflect on the past

One aspect of successful impact is when you consistently deliver against all and likely exceed some expectations for your core priorities. Capture what you've accomplished and learned since your last Connect.

Summarize your impact

Core priorities

What impact did you have for each of your core priorities?

Coding work:

- **Added query with time Series(historised data) May 23**
 - Modified the DMM API to enable users to query through time Stamp. This will enable users to use the ability of historization feature of ADX and to do more Indepth ROC between timeframe. Also made sure this feature could be used alongside with other filters.
 - Added sort by feature and no of records in the query API.
 - Added the L0 test for all the code.
- **Multihop query June 23**
 - Added mutihop feature to related data API to enable users to query nodes that are not directly related to the source and also find path and relationships to it. Made the distance/ hops from source node configurable in API.
 - In this API enabled user to filter through fields stored in both cosmos and azure data explorer.
 - Enhanced this API to enable users to add path filters. Though this feature it will enable users to provide the entity name that should be present between source and target.
- **Consumption and performance benchmarking July 23**
 - Performed benchmarking both ingestion and consumption APIs for all the resources like ADT, ADX and cosmos. Also highlighted when the performance of these resources was peaking and which resource was more costly. This further helped in the decision to remove ADT.
 - Optimized the ingestion parameters like RequestBatchSize, PollingInterval, partitionCount etc for optimal performance.
 - Modified the config files to produce the data to comply with validation changes. Made sure that there is no validation error among all 330k nodes and 450k relationships.
 - Wrote python script for testing the API with parallel calls and finding latency
 - Summarized all the insights and findings in a comprehensive document for future reference.
- **Redis Bicep changes and code changes Aug**
 - Wrote bicep files to generate EventHub and Redis during our deployment. Figured out the permission required by other resource to communicate with EventHub and Redis and automated it in bicep.
 - Removed the dependency from using connection string by modifying the SKG and ingestion codebase to fetch the keys dynamically through managed identity. This will decrease the risk of password leak as the connection string is not stored anywhere.
 - Increased the security by disabling public access of Redis and making everything work under virtual network.
- **Bring your own store Aug - Sept**
 - Modified the Bicep to allow users to bring their own Storage, Cosmos, Azure DataExplorer. Through this they can use the same data store for multiple purpose which would increase their savings.
 - Configured the existing virtual network and private endpoint to be able to read and write in external resources.
- **Enhancing Inbuilt entities and sample data Sept 23**
 - Modified the generate sample data project to comply with the new validations introduced in last semester. This help in testing our solution with greater no of entities.
 - Updated the already present inbuilt entities with latest version and modified the ingestion codebase to work with the same. Added entities related to work operation and job classes. This gives the users to use most updated and latest entities.
 - Modified the sample data project to work for stream ingestion by limiting the size of JSON produced below 1MB. Though this stream ingestion testing could be done for large no of dataset.
- **Hackathon/ partner work Oct 23**
 - Developed python script to map data from customer data to ISA 95 schema.
 - Worked exhaustively with PM team to fine tune the mapping, changed the logic to speed up the code and decrease number of relationships.
 - Ingested the data in our instance and debugged the missing data and relationships.
- **Azure Policies Nov 23**
 - Wrote azure policies in the azure portal to deny creation of AKS resource that do not adhere to our OS SKU requirements at subscription level.
 - This will help us be S360 compliant as latest AKS Mariner require all the OS SKU of all the agent pool to be of azure Linux type.
- **Removing List Keys API Nov 23**
 - Updated the consumption codebase to use managed identity for connection between cosmos and ADX while querying in ADX.
 - Updated the code to use managed identify credential for connection between the service and Redis resource.

- With these changes there will be no need for list keys API which will increase security and would comply with latest changes of RP team

POC:

- **Fibric spikes** *Oct 23*
 - Explored how to connect the function app to Onelake in Microsoft fabric. Used service principal for authentication. Wrote a dummy function app to read and write data in Onelake.
 - Explored shortcut feature of Onelake. Shortcut feature is used to connect to external storage and fetch data.
 - Used function app to poll the files in Onelake at a folder level. This was done as an alternate method to mimic blob triggers as they are currently not supported in Onelake
- **Removing List Keys API** *Oct 23*
 - Did POCs for using managed identity in the query API to connect to cosmos from ADX. Earlier we were using the connection string which had to be fetched from list keys API. With my POC we can use the bearer token to authenticate which improves security and would be compatible with UMI changes.
 - Did POC for using managed identity for Redis. Earlier we used connection string for Redis which required the list keys API. With my POC we can do that with system managed identity. I explored the required permissions and corresponding bicep changes needed for the connection.

Bug Fixes:

- **Managed private Endpoint for storage**
 - Added managed private endpoint from Azure data explorer to BJS storage. Automated the creation and approval of the endpoint in bicep.
 - It was added in order to establish the connection when the public Access of the resources is disabled.
- **Function app basic authentication** *May23*
 - Fixed the bicep template to incorporate the latest change by function app team.
 - Added PowerShell code to enable Basic authentication and push the code to function app.
- **URL encoding**
 - Added URL encoding for the API responses containing URL.

Diversity & Inclusion (D&I) Core Priority

What impact did your actions have in contributing to a more diverse and inclusive Microsoft?

- Organized the inter MCI indoor games competition with few folks from the energy team for the give event. This event gave opportunity to interact with other members of our sister team and learn new culture from them
- Ramping up new hires. I worked with Dhruv closely and helped him in understanding our codebase and technologies we use. I tried to make sure he has a friendly environment,
- Worked with teams from different time zones during partner work. I incorporated their office time whenever I had to setup meetings.



Manager comments

Devesh, it's a solid 6months of heavy hitting delivery from your side which was crucial in the success of SKG.

- You demonstrated good coding skills and consistently takes ownership of tasks exhibiting topnotch accountability.
- Hunger for work and always interested in picking up challenging tasks, delivering them on time and with perfection.
- Quickly ramped up on the data mapper tool, enhanced it heavily with fixes, additional capabilities and the contributions for IPG Hackathon and ingestion process was critical in success of Ingestion workstream and also the overall hackathon. commitment and excellence shown during the 3-week Hackathon period is just a testament of passion towards work and delivering it the right way.
- Also, for the June Perf/Cost benchmarking was done thoroughly and very quick in writing utilities that helped during that process.
- Voluntarily picking up infra work and up-skilling was quite commendable.
- Acknowledged on all of the feature contributions above.

Reflect on a challenge or setback

In addition to your accomplishments relative to your core priorities, you must demonstrate a growth mindset to deliver successful impact. Share how you demonstrated a growth mindset by taking personal accountability for setbacks, asking for feedback, and applying learnings to have greater impact.

Consider when you could have done something differently. How will you apply what you learned to make an even greater impact?

- For Bring your own store a better communication with the RP team could be established. The PR was raised multiple times and reverted few times to incorporate the changes in the current release.
- In the partner related work required a lot of small mapping changes. This increased the time as each change needed to be validated and checked.



Manager comments

Agreed and it's a perfect reflection.

Plan for the future

Think about opportunities for impact in the upcoming Connect period. What are your 3-5 core priorities?

Review your core priorities

Core priorities

1. Migrating to Microsoft Fabric

Focus areas for the upcoming period to drive business impact:

In upcoming semester we will be migrating to Microsoft fabric. We will shift most of the resources containing data to fabric for a better security and inter connection between the resources.

Success measures:

Moving to fabric will allow the users to have a better control over the data. The user can also use power bi tools on the data. Further as all data is at one place(even from different Microsoft services), it will lead to better insights and faster root cause analysis.

2. Reducing dependency from Azure Digital twin

Focus areas for the upcoming period to drive business impact:

In the last benchmarking and performance calculation we saw that there was a lot of 429 error(Too many requests) that azure digital twin was throwing. This resulted to throttling issues. This semester we are planning to remove azure digital twin and to ingest data directly to azure data explorer. This will give us better control on the DTDLs also as we can self validate the DTDL.

Success measures:

Using direct ingestion we will see a direct improvement in ingestion benchmarking results. There will be significant reduction in retries.

3. Getting familiarize with Copilot Codebase

Focus areas for the upcoming period to drive business impact:

- Copilot team is our sister team and our work is closely coupled. This semester I will try to look deeper in their code base to get myself familiar with their design.
- This will enable me to switch the work seamlessly between SKG and copilot and would also help me in growing my knowledge in copilot working and in generative AI.
- Contribute towards any partner hackathons.

Success measures:

- Ramping up in CoPilot work quickly and seamlessly.
- Better overall design keeping in mind both the steams.

4. **Better Design Patterns****Focus areas for the upcoming period to drive business impact:**

- I will try to incorporate the best design patterns in the SKG codebase.
- Study other codebases on how they implement the similar work and try to bring the best coding practices in SKG.
- Enrich myself on latest design patterns through online courses.
- Review already implemented and ongoing design of SKG features and bring improvements in it.

Success measures:

- Better overall design of the DMM product.
- Easy maintainability and debugging of the DMM product.

Diversity & Inclusion (D&I) Core Priority

To realize our mission to empower every person and organization on the planet to achieve more, each and every one of us must help create a diverse and inclusive culture where everyone can bring their full and authentic self, where all voices are heard, and where we do our best work as a result. Consider how you will **deliver successful impact by consistently demonstrating Microsoft's cultural expectations and values.**

Select and commit to at least one focus area from the options below, or create one of your own.

- Seek diverse perspectives and incorporate feedback, apply insights from a variety of sources and people
- Mentor and/or support a person who may benefit

How will you define success as you activate your D&I Core Priority?

- The perspective and feedback from others would be a good measure for success.

**Manager comments**

Also, would like to see you taking opportunities to enhance design skills and learning generative AI advancements and capabilities.

Identify your opportunities for growth

We are all expected to embrace a growth mindset and focus on continuous learning to deliver successful impact.

What 1-2 actions will you take to focus on learning, growth, and development over the next period?

Learn more about Microsoft fabric. Especially how it handles data through different stages.
Apply better coding standards in my pull request.
Better mentor for new hires.

What support do you need to fulfill these actions? Fill in any that apply.

Opportunities to (optional):

No Response

Perspective or coaching on (optional):

No Response

Learning on (optional):

No Response



Manager comments

Also actively take part in recruitment process and mentoring new hires/interns.

Conversation starters (optional)

Choose any of these optional prompts to continue great conversations with your manager. Write a brief response (1-15 words).

Of my strengths, the ones I'd like to apply more in my work are:

In my work, I feel motivated or energized by:

In my work, I lose motivation or energy from:

I know I can't do everything on my own. I'd like help with:

It can be challenging to prioritize important topics for 1:1 conversations. Plan ahead and flag any of the following topics you'd like to discuss with your manager in upcoming conversations.

Career: Discuss ways to grow my career to align with my strengths, interest & aspirations.

Recognition: Discuss my [recognition preferences](#) and highlight my contributions.

Opportunities: Increase my responsibilities, such as new core priorities, stretch assignments, promotion or becoming a people manager.

Alignment: Clarify priorities and expectations for successful impact.

Learning: Explore potential development and growth opportunities to optimize your performance.

Feedback: Get input on how I'm doing in my role and how to deliver great impact.

Flexibility: Consider how you work best and discuss your workplace preferences (i.e. site, location, hours)

Wellbeing: Discuss any dimensions of being well: mental, emotional, physical, financial.

Check out aka.ms/career for resources and guides on developing your career.



Manager comments